UNIT - I
Methods of Design of Concrete Structures

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Methods of Design - Overview

- **Reinforced concrete (RC)** = concrete (strong in compression) + steel (strong in tension), acting together because of **bond**, equal thermal expansion and cover protection.
- Aim of structural design: the structure must be **safe** (strength, stability), **serviceable** (deflection, cracking, vibration), **durable** and **economical**.
- Code used: **IS 456:2000** (Plain and Reinforced Concrete - Code of Practice). Loads: IS 875, seismic: IS 1893. Design aids: SP 16.

Elastic method (WSM) → Ultimate load method (ULM) → Limit state method (LSM)

Method	Basis	Safety provided by
Elastic / Working stress	Linear stress-strain, working loads	Permissible stresses (factor of safety on materials)
Ultimate load	Collapse load, non-linear stress block	Load factor on working loads
Limit state	Probability, limit states	Partial safety factors on loads AND materials

Concept of Elastic Method (Working Stress Method)

- Materials behave **elastically**; stress is proportional to strain up to working loads.
- Stresses under **working (service) loads** are kept below **permissible stresses** = (strength / factor of safety).

- Concrete: permissible stress about 1/3 of cube strength; steel: about 0.55 fy.
- Uses **modular ratio m** to convert steel to an equivalent area of concrete (transformed section).
- Drawback: ignores non-linear behaviour, creep, shrinkage; no idea of real margin against failure.

Concept of Ultimate Load Method (ULM)

- Also called **load factor method**. Working loads are multiplied by a **load factor** (1.5 - 2.0) to get ultimate loads, and the section is designed to just fail at that load.
- Uses the **actual non-linear** stress-strain behaviour of concrete at failure (rectangular / parabolic stress block).
- Ultimate moment (rectangular block): $M_u = A_{st} f_y (d - A_{st} f_y / (1.7 f_{ck} b))$
- Advantages: more realistic, economical sections, rational. Drawback: **serviceability (deflection, cracks) not checked** - slender sections.

Concept of Limit State Method (LSM)

- A **limit state** is the condition beyond which the structure ceases to fulfil its intended function.
- LSM considers **both** safety (collapse) and serviceability together, using **partial safety factors** for loads and materials. It is the **main method of IS 456:2000**; WSM is retained as an alternative (Annex B).

Summary: WSM ⟶ stresses at working load. ULM ⟶ strength at ultimate load. LSM ⟶ collapse + serviceability with partial safety factors (best of both).

2. Working Stress Method as per IS 456 (Annex B)

Assumptions (B-1.3)

- At any cross-section, **plane sections remain plane** after bending.
- All tensile stresses are taken by **steel only** - concrete in tension is neglected.
- Stress is proportional to strain in concrete (**linear elastic**) and in steel.
- The **modular ratio** $m = E_s / E_c$ is the same for steel in tension and compression (equivalent area of steel = $m\ A_s$).
- Perfect **bond** between steel and concrete.

Modular Ratio and Material Constants

$$m = 280 / (3\ \sigma_{cbc})$$

- σ_{cbc} = permissible compressive stress in concrete in bending. (Here $E_c = 5000\ \sqrt{f_{ck}}$, $E_s = 2 \times 10^5\ N/mm^2$.)
- Permissible stresses (IS 456 Tables 21, 22 - flexure):

Grade	σ_{cbc} (bending)	σ_{cc} (direct)	$m = 280/(3\sigma_{cbc})$
M20	7.0	5.0	13.33
M25	8.5	6.0	10.98
M30	10.0	8.0	9.33
M35	11.5	9.0	8.12
M40	13.0	10.0	7.18
Steel	Fe 250	Fe 415	Fe 500
σ_{st} (N/mm^2)	140 (up to 20 mm dia), 130 (above)	230	275

Permissible shear stress and bond stress are also taken from IS 456 Tables 23 and 21 (bond stress for plain bars in M20 = 0.8 N/mm^2; increased by 60% for deformed bars).

Behaviour of a Singly Reinforced Section (WSM)

- Let b = width, d = effective depth, x = depth of neutral axis (NA), A_{st} = tension steel.
- **NA is located** by taking moment of areas about the NA: $b\,x^2/2 = m\,A_{st}\,(d - x)$ for the cracked section.
- Critical (balanced) NA: concrete and steel reach permissible stresses **simultaneously**.

$$k = x_c/d = m\,\sigma_{cbc} / (m\,\sigma_{cbc} + \sigma_{st})$$

$$j = 1 - k/3 \text{ (lever arm } z = j\,d)$$

- Moment of resistance (concrete side): $M_c = (1/2)\,\sigma_{cbc}\,k\,j\,b\,d^2 = Q\,b\,d^2$, where $Q = 0.5\,\sigma_{cbc}\,k\,j$.
- Moment of resistance (steel side): $M_s = \sigma_{st}\,A_{st}\,j\,d$.
- Percentage of steel for balanced section: $p_{t,bal} = 50\,k\,\sigma_{cbc} / \sigma_{st}$ (%).

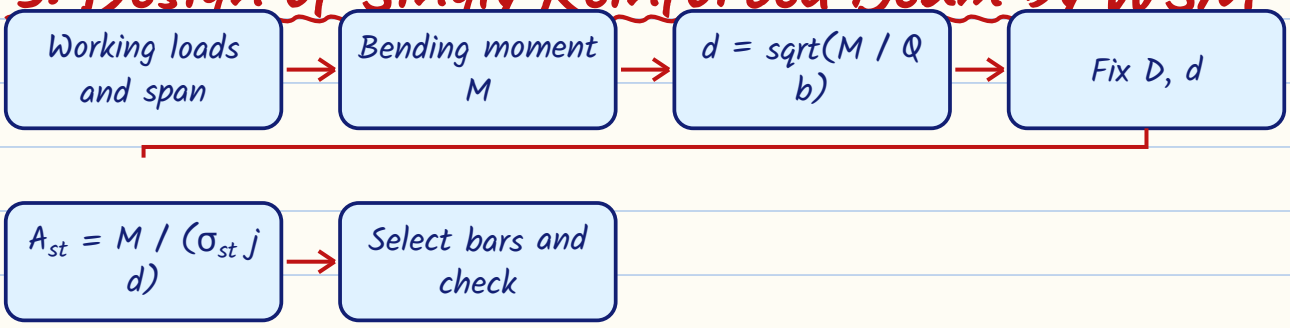
M20 with	k	j	Q (N/mm^2)	$p_{t,bal}$ %
Fe 250 (σ_{st}=140)	0.400	0.867	1.21	1.00
Fe 415 (σ_{st}=230)	0.289	0.904	0.913	0.439
Fe 500 (σ_{st}=275)	0.253	0.916	0.812	0.323

Types of Section (WSM)

- **Balanced** (critical): $x = x_c$. Steel and concrete reach limits together; $A_{st} = A_{st,bal}$.
- **Under-reinforced**: A_{st} < balanced, $x < x_c$. **Steel yields first** ⟶ ductile, warning before failure; $M = \sigma_{st}\,A_{st}\,j\,d$.
- **Over-reinforced**: A_{st} > balanced, $x > x_c$. **Concrete crushes first** (brittle, sudden). In WSM, $M = (1/2)\,\sigma_{cbc}\,b\,x\,(d - x/3)$ governed by concrete.

IS 456 (WSM): the section may be under- or over-reinforced; under-reinforced preferred, because of ductile failure and warning. In LSM, over-reinforced sections are NOT permitted.

3. Design of Singly Reinforced Beam by WSM



Design Steps

- 1. Assume b (about D/2 to D/3) and grade of concrete and steel; get σ_{cbc}, σ_{st}, m.
- 2. Compute k, j, Q for the balanced section.
- 3. Working moment $M = w\,l^2 / 8$ (simply supported beam, UDL including self weight).
- 4. Required effective depth $d = \sqrt{M / (Q\,b)}$; adopt a **greater** round value; D = d + cover + half bar dia.
- 5. $A_{st} = M / (\sigma_{st}\,j\,d)$; provide bars and check $A_{st,min}$, spacing, and bond.
- 6. Check shear and deflection (span/depth). Provide stirrups.

Worked Example 1 (WSM)

Q. Design a simply supported RC beam of effective span 5 m carrying a total working load of 15 kN/m (including self weight). Use M20 and Fe 415. Take b = 250 mm.

- $\sigma_{cbc} = 7$, $\sigma_{st} = 230$, $m = 13.33$. $k = 93.33/(93.33+230) = 0.289$, $j = 1 - 0.289/3 = 0.904$, $Q = 0.913$.
- $M = w\,l^2/8 = 15 \times 5^2 / 8 =$ **46.875 kN.m**
- $d = \sqrt{46.875 \times 10^6 / (0.913 \times 250)} = \sqrt{205{,}367} =$ **453 mm**.

Adopt d = 455 mm, overall depth D = 500 mm (effective cover 45 mm).

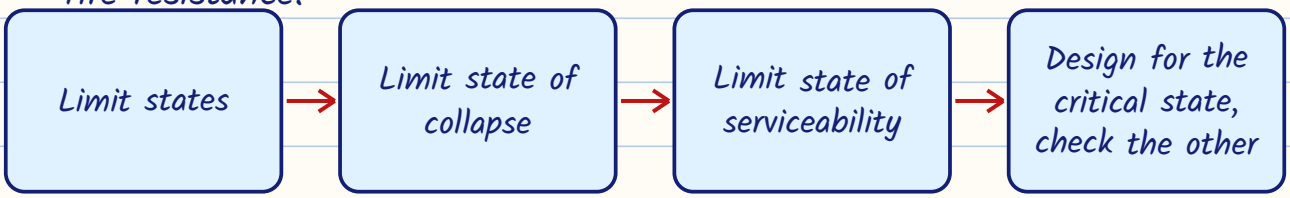
- A_{st} = 46.875 x 10^6 / (230 x 0.904 x 455) = **496 mm^2**. Provide 3 - 16 mm dia bars (603 mm^2).
- Check actual NA: 125 x^2 = 13.33 x 603 (455 - x) ⟶ x = 141.9 mm > x_c = 0.289 x 455 = 131.4 mm. The section is slightly over-reinforced, concrete governs.
- Actual M = 0.5 x 7 x 141.9 x 250 x (455 - 141.9/3) = **50.6 kN.m > 46.9 kN.m** ⟶ safe.

Final: b x D = 250 x 500, 3 - 16φ at bottom, 2 - 12φ hanger bars at top, 8 mm 2-legged stirrups (spacing from shear design).

4. Limit State Philosophy as per IS 456

Limit States

- **Limit state of collapse (strength)**: flexure, compression, shear, torsion; overturning, sliding, buckling, rupture.
- **Limit state of serviceability**: **deflection**, **cracking**, durability, vibration, fire resistance.



Characteristic Loads and Strengths

- **Characteristic strength of concrete f_{ck}** = 28-day cube strength below which not more than **5%** of test results fall.
- **Characteristic strength of steel f_y** = yield (0.2% proof) stress with 5% defective probability.
- **Characteristic load** = load with not more than 5% probability of being exceeded during the design life.

- **Design value** = characteristic value / partial safety factor (for materials) or characteristic value x factor (for loads).

Partial Safety Factors

Material	γ_m	Design strength
Concrete	1.5	$f_d = 0.446\ f_{ck}$ (= $0.67\ f_{ck}/1.5$)
Steel	1.15	$f_{sd} = 0.87\ f_y$ (= $f_y/1.15$)

Partial Safety Factors for Loads (IS 456 Table 18)

Load combination	DL	IL	WL / EL
DL + IL	1.5	1.5	-
DL + WL (or EL)	1.5 or 0.9*	-	1.5
DL + IL + WL (or EL)	1.2	1.2	1.2

- *0.9 DL is used when stability against overturning or stress reversal is critical.
- Design load (**factored load**) = γ_f x characteristic load. Eg. w_u = 1.5 (DL + LL).

Assumptions in Flexure (IS 456 Cl. 38.1)

- Plane sections remain plane (strain varies linearly).
- **Maximum strain in concrete** at the extreme compression fibre = **0.0035** in bending.
- Stress-strain curve of concrete: **parabolic up to strain 0.002, then constant** at 0.446 f_{ck} (design).
- **Tensile strength of concrete is ignored.**
- Stress in steel from the design stress-strain curve (Fe 250: 0.87 f_y after yield; Fe 415/500: non-linear curve).
- Max strain in tension steel at failure must be at least **$(0.87\ f_y/E_s)$ + 0.002** (ductility), which restricts x_u to $x_{u,max}$.

Equivalent Rectangular Stress Block

- Parabolic-rectangular area is replaced by an equivalent rectangle:

Compression $C = 0.36\ f_{ck}\ b\ x_u$

Centroid of C from top = $0.42\ x_u$ (lever arm $z = d - 0.42\ x_u$)

Tension $T = 0.87\ f_y\ A_{st}$

5. Advantages of Limit State Method over Other Methods

WSM	LSM
Linear elastic assumption not true near failure	Considers real non-linear behaviour
Single factor of safety on materials only	Separate partial factors on loads and materials
Factor of safety against collapse is not known	Reliable margin against collapse is known
Serviceability not considered directly	Checks both strength and serviceability
Ignores creep, shrinkage, redistribution	Allows moment redistribution and better detailing
Uneconomical (heavier sections)	Economical, about 10-15% saving of steel
Same safety for DL and LL	Different factors reflect uncertainty of each load

6. Analysis of Singly Reinforced Rectangular Beam by LSM

C = T equilibrium → Find x_u → Compare with $x_{u,max}$ → Find M_u = T x lever arm

Neutral Axis Depth

$0.36\ f_{ck}\ b\ x_u = 0.87\ f_y\ A_{st} \longrightarrow x_u = 0.87\ f_y\ A_{st} / (0.36\ f_{ck}\ b)$

Limiting Depth of Neutral Axis (IS 456 Cl. 38.1, Table in Annex G)

- At the **limiting** (balanced) condition, concrete strain = 0.0035 and steel strain = $0.87 f_y/E_s + 0.002$.

$$x_{u,max}/d = 0.0035 / (0.0055 + 0.87 f_y/E_s)$$

Steel	$x_{u,max}/d$	$M_{u,lim}/(f_{ck} b d^2)$	$p_{t,lim}$ (M20) %
Fe 250	0.531	0.148	1.76
Fe 415	0.479	0.138	0.96
Fe 500	0.456	0.133	0.76
Fe 550	0.440	0.129	0.66

Types of Section in LSM

- **Under-reinforced**: $x_u < x_{u,max}$ (steel yields first; ductile). Preferred. $M_u = 0.87 f_y A_{st} (d - 0.42 x_u)$.
- **Balanced**: $x_u = x_{u,max}$. $M_{u,lim} = 0.36 f_{ck} b x_{u,max} (d - 0.42 x_{u,max})$.
- **Over-reinforced**: $x_u > x_{u,max}$. **Not permitted**; take $M_u = M_{u,lim}$ (steel does not yield, brittle failure).

Moment of Resistance

$$M_u = 0.87 f_y A_{st} d (1 - A_{st} f_y / (b d f_{ck}))$$

$$M_{u,lim} = 0.138 f_{ck} b d^2 \text{ (Fe 415) } 0.148 \text{ (Fe 250) } 0.133 \text{ (Fe 500)}$$

- Using percentage of steel $p_t = 100 A_{st}/(b d)$:

$$M_u / (b d^2) = 0.87 f_y (p_t/100) (1 - f_y p_t / (100 f_{ck}))$$

Design Formula for A_{st} (when M_{u} <= M_{u,lim})

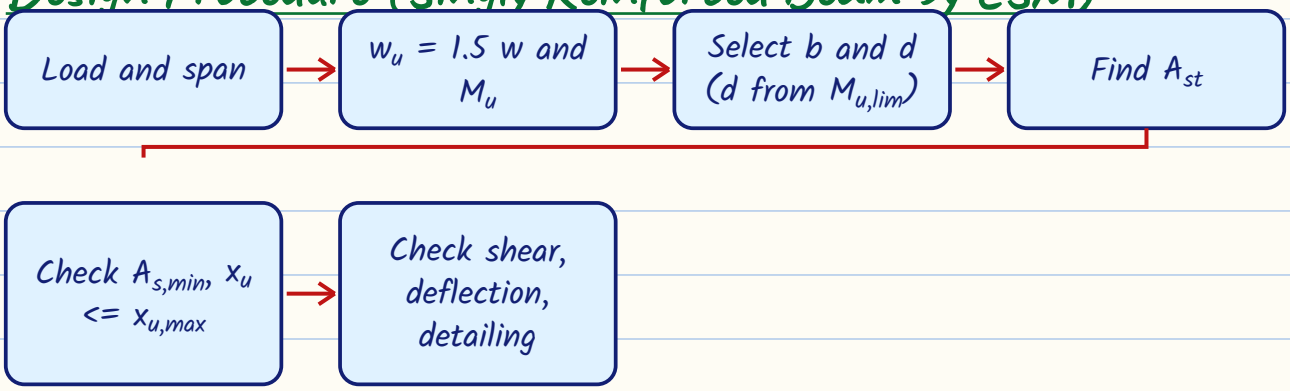
$$A_{st} = (0.5 f_{ck} / f_y) [1 - \text{sqrt}(1 - 4.6 M_u / (f_{ck} b d^2))] b d$$

Limits on Steel (IS 456 Cl. 26.5.1)

- **Minimum tension steel**: $A_{s,min} = 0.85 b d / f_y$ (Fe 415: 0.205% of bd).

- **Maximum tension steel**: 0.04 b D (4% of gross area).
- Side face steel 0.1% of web area if D > 750 mm (spaced not more than 300 mm).

Design Procedure (Singly Reinforced Beam by LSM)



Worked Example 2 (SRS beam by LSM)

Q. A simply supported beam of effective span 6 m carries imposed load 12 kN/m and dead load 8 kN/m (excluding self weight). Design for flexure. Use M20, Fe 415. Take b = 250 mm, D = 550 mm, clear cover 30 mm, using 22 mm bars and 8 mm stirrups.

- Effective depth d = 550 - 30 - 10 (approx half bar plus stirrup adjustment) = **510 mm**. Self weight = 0.25 x 0.55 x 25 = 3.44 kN/m.
- Total w = 8 + 12 + 3.44 = 23.44 kN/m ⟶ w_u = 1.5 x 23.44 = **35.16 kN/m**.
- $M_u = w_u l^2/8 = 35.16 \times 6^2/8$ = **158.2 kN.m**
- $M_{u,lim} = 0.138 \times 20 \times 250 \times 510^2$ = **179.5 kN.m** > 158.2 kN.m ⟶ singly reinforced section is adequate.
- $M_u/(b\,d^2) = 158.2 \times 10^6 / (250 \times 510^2) = 2.433$ N/mm²; 4.6 x 2.433 / 20 = 0.5596 ⟶ sqrt(1 - 0.5596) = 0.6636
- $A_{st} = (0.5 \times 20 / 415) \times (1 - 0.6636) \times 250 \times 510$ = **1033 mm²**
- Provide **3 - 22φ (1140 mm²)**. p_t = 1140/(250 x 510) = 0.894%. $A_{s,min}$ = 0.85 x 250 x 510 / 415 = 261 mm² <1140 OK, < 4% bD.
- Check $x_u = 0.87 \times 415 \times 1140 / (0.36 \times 20 \times 250)$ = **228.7 mm** < $x_{u,max}$ = 0.48 x 510 = **244.8 mm** ⟶ under-reinforced OK.

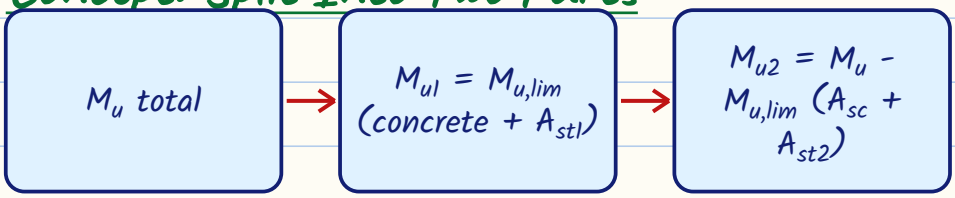
- Actual M_u = 0.87 x 415 x 1140 x (510 - 0.42 x 228.7) = **170.4 kN.m** > **158.2** ⟶ safe.
- Spacing: (250 - 2 x 30 - 2 x 8 - 3 x 22)/2 = 54 mm > 25 mm and > bar dia OK. Span/depth = 6000/510 = 11.8 < 20 OK.

Final: 250 x 550, 3 - 22φ at bottom, 2 - 12φ at top, 8φ stirrups.

7. Doubly Reinforced Rectangular Beams

- Used when **M_u > $M_{u,lim}$** and section size can not be increased (architectural or head-room limit), or when moment reversal occurs.
- Steel is also provided in the **compression zone** (A_{sc}) in addition to tension steel (A_{st}).
- Advantages: increases moment capacity, **ductility**, reduces long-term deflection (compression steel resists creep), helps stirrup support.

Concept: Split Into Two Parts



- x_u is taken equal to **$x_{u,max}$** for the design case (so steel yields).

$$M_{u1} = M_{u,lim} = 0.36\, f_{ck}\, b\, x_{u,max}\, (d - 0.42\, x_{u,max})$$

$$M_{u2} = M_u - M_{u,lim} = A_{sc}\, (f_{sc} - f_{cc})\, (d - d')$$

- f_{cc} = compressive stress in concrete at the level of compression steel = **0.446 f_{ck}** (to avoid double counting of the concrete area).

$$A_{sc} = M_{u2} / [\, (f_{sc} - 0.446\, f_{ck})\, (d - d')\,]$$

$$A_{st1} = 0.36\, f_{ck}\, b\, x_{u,max} / (0.87\, f_y)$$

$$A_{st2} = A_{sc} (f_{sc} - f_{cc}) / (0.87 f_y) \quad A_{st} = A_{st1} + A_{st2}$$

Stress in Compression Steel (fsc)

- Strain at the level of A_{sc}: $\varepsilon_{sc} = 0.0035 (1 - d'/x_{u,max})$; f_{sc} is read from the stress-strain curve (SP 16 Table).

d'/d	0.05	0.10	0.15	0.20
f_{sc} Fe 250	217	217	217	217
f_{sc} Fe 415	355	353	342	329
f_{sc} Fe 500	424	412	395	370

- For Fe 250: $f_{sc} = 0.87 f_y = 217.5$ N/mm^2. d' = distance from the compression face to centroid of A_{sc} (effective cover).

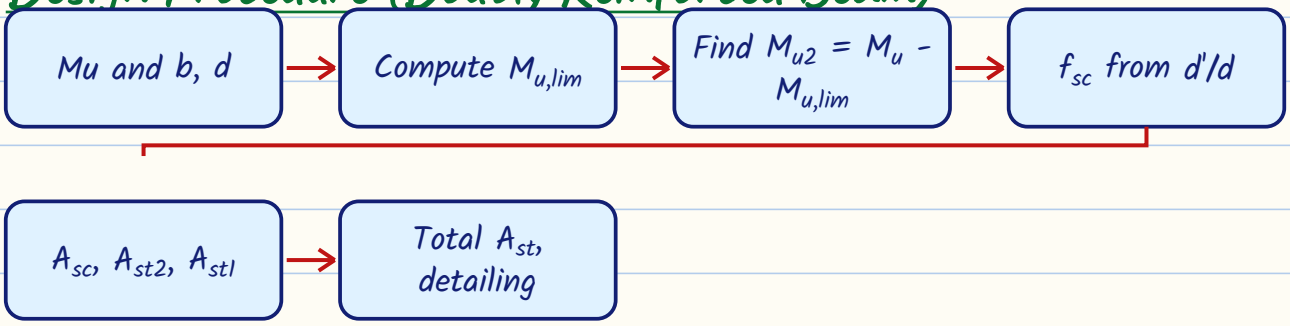
Analysis of a Given Doubly Reinforced Section

- Assume x_u; find ε_{sc}, f_{sc}; solve equilibrium C = T:

$$0.36 f_{ck} b x_u + A_{sc} (f_{sc} - f_{cc}) = 0.87 f_y A_{st}$$

- If $x_u <= x_{u,max}$: $M_u = 0.36 f_{ck} b x_u (d - 0.42 x_u) + A_{sc} (f_{sc} - f_{cc}) (d - d')$.
- If $x_u > x_{u,max}$: take $x_u = x_{u,max}$ (limit) and compute the same expression.

Design Procedure (Doubly Reinforced Beam)



- Provide **stirrups** (closely spaced) around compression bars to prevent buckling: spacing <= 16 x dia of compression bar, <= 300 mm.

Worked Example 3 (Doubly Reinforced Beam)

Q. A beam 250 mm x 500 mm (effective cover 50 mm both sides) has

factored moment M_u = 230 kN.m. Design the reinforcement. Use M20 and Fe 415.

- d = 500 − 50 = **450 mm**, d' = 50 mm, d'/d = 0.111.
- $M_{u,lim}$ = 0.138 × 20 × 250 × 450^2 = **139.7 kN.m** < 230 ⟶ **doubly reinforced** section needed.
- M_{u2} = 230 − 139.7 = **90.3 kN.m**
- $x_{u,max}$ = 0.48 × 450 = 216 mm. ε_{sc} = 0.0035 (1 − 50/216) = 0.00269 ⟶ f_{sc} = **351 N/mm²** (from table, by interpolation of d'/d = 0.111).
- f_{cc} = 0.446 × 20 = 8.92 N/mm²
- A_{sc} = 90.3 × 10^6 / ((351 − 8.92)(450 − 50)) = 90.3 × 10^6 / 136,832 = **660 mm²** ⟶ 3 − 18φ (763 mm²).
- A_{st1} = 0.36 × 20 × 250 × 216 / (0.87 × 415) = 388,800 / 361.05 = **1077 mm²**
- A_{st2} = A_{sc} (f_{sc} − f_{cc})/(0.87 f_y) = 660 × 342.08 / 361.05 = **625 mm²**
- A_{st} = 1077 + 625 = **1702 mm²** ⟶ provide 6 − 20φ (1885 mm²) in two layers (3 + 3).
- Check max steel: 4% of bD = 0.04 × 250 × 500 = 5000 mm² > 1885 OK; A_{sc} = 763 < 5000 OK.

Result: tension steel 6 − 20φ (two layers), compression steel 3 − 18φ, stirrups 8φ at close spacing. Always check d'/d and $x_{u,max}$.

8. Detailing Requirements (IS 456 Cl. 26)

Cover to Reinforcement (Table 16, beams/slabs)

Exposure	Nominal cover (mm)
Mild	20
Moderate	30
Severe	45
Very severe	50

Extreme	75

- Cover must be at least bar dia and ensures **fire resistance, corrosion protection, bond**.

Spacing of Bars

- Minimum horizontal clear spacing: **the larger of** bar diameter, **25 mm**, or (maximum aggregate size + 5 mm).
- Vertical spacing between layers: >= 15 mm, 2/3 of maximum aggregate size, or bar dia.
- **Maximum clear spacing** in tension zone (Table 15): Fe 250: 300 mm; Fe 415: 180 mm; Fe 500: 150 mm (for no redistribution).

Effective Span, Depth and Control of Deflection

- **Effective span** of simply supported beam = lesser of (clear span + d) and (centre to centre of supports).
- **Effective depth d** = D - effective cover (cover + stirrup + half bar dia).
- **Basic span/depth ratios** (Cl. 23.2.1): cantilever **7**, simply supported **20**, continuous **26** (for span up to 10 m). Spans > 10 m: multiply by 10/span in metres (not for cantilevers).
- Modification factors: for **tension steel** (Fig. 4: depends on $f_s = 0.58\ f_y\ A_{st,req}/A_{st,prov}$ and p_t), **compression steel** (Fig. 5), **flanged beams** (Fig. 6).

$$(l/d)_{max} = (\text{basic } l/d) \times k_t \times k_c \times k_f$$

- Anchorage of tension bars: $L_d = \varphi\ \sigma_s / (4\ \tau_{bd})$; continued past the support by L_d.

Remember (beam): Ast,min = 0.85 bd/fy ; Ast,max = 0.04 bD ; cover 20-75 ; l/d = 7, 20, 26.

9. Quick Revision

- WSM: elastic, permissible stress, $m = 280/(3\,\sigma cbc)$, $M = Q\,b\,d^2$, $Ast = M/(\sigma st\,j\,d)$.
- ULM: load factor 1.5 – 2, non-linear block; no serviceability check.
- LSM: $\gamma c = 1.5$, $\gamma s = 1.15$, loads 1.5 / 1.2; 0.446 fck, 0.87 fy; $C = 0.36\,fck\,b\,xu$; $z = d - 0.42\,xu$.
- xu,max/d: **0.53** / **0.48** / **0.46** / **0.44** for Fe 250 / 415 / 500 / 550.
- Mu,lim = 0.148 / 0.138 / 0.133 / 0.129 x fck b d^2.
- DRS: $Mu2 = Asc\,(fsc - fcc)(d - d')$, $fcc = 0.446\,fck$, $Ast = Ast1 + Ast2$.

Important Questions

- 1. Explain the concepts of working stress, ultimate load and limit state methods. Compare them.
- 2. List the assumptions of WSM. Derive k, j and Q for a balanced singly reinforced section.
- 3. Define balanced, under-reinforced and over-reinforced sections. Which is preferred and why?
- 4. Design a singly reinforced beam by WSM for given span and load (M20, Fe 415).
- 5. Explain limit state philosophy. State characteristic strength, load and partial safety factors.
- 6. Derive the stress block parameters 0.36 fck b xu and 0.42 xu. Derive $x_{u,max}/d$ for Fe 415.
- 7. Derive $M_{u,lim}$ for a singly reinforced rectangular beam. Why is the over-reinforced section not permitted?
- 8. Write the advantages of LSM over WSM.
- 9. Design a doubly reinforced beam for the given M_u. Explain how f_{sc} is obtained.